

# A proof of the conjectured linear recurrence and generating function for OEIS A186460

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## 1 The conjecture

The Online Encyclopedia of Integer Sequences records [A186460](#) as

*Number of  $(n+1) \times 8$  binary arrays with every  $2 \times 2$  subblock commuting with each of its horizontal and vertical  $2 \times 2$  subblock neighbors*

and carries in its Formula section the lines:

Empirical:  $a(n) = 2a(n-1) - 2a(n-3) + 2a(n-4) - a(n-5)$  for  $n > 13$ .

Empirical g.f.:  $x(245 - 198x + 6x^2 + 255x^3 - 115x^4 - 144x^5 + 369x^6 - 547x^7 - 36x^8 - 640x^9 + 484x^{10} - 300x^{11} + 784x^{12}) / ((1 - x)(1 - x - x^2 + x^3 - x^4))$ . - Colin Barker, Apr 18 2018

That line carries no separate signature; the entry itself is by R. H. Hardin (Feb 22 2011).

The line quoted ending in that signature is contributed by Colin Barker on Apr 18 2018.

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 9, last modified Apr 18 2018, the entry records no proof and links to none, so the statement is open. This note settles it.

Written out, the claim is

$$a(n) = 2a(n-1) - 2a(n-3) + 2a(n-4) - a(n-5).$$

stated for  $n > 13$ .

Written out, the claim is

$$\sum_{n \geq 1} a(n) x^n = \frac{245x - 198x^2 + 6x^3 + 255x^4 - 115x^5 - 144x^6 + 369x^7 - 547x^8 - 36x^9 - 640x^{10} + 484x^{11} - 300x^{12}}{1 - 2x + 2x^3 - 2x^4 + x^5}$$

## 2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let  $A$  be an  $n' \times 8$  array over  $\{0, \dots, 1\}$  with  $n' = n + 1$  rows.

Write  $B_{i,j}$  for the  $2 \times 2$  window with upper left corner  $(i, j)$ . Its horizontal neighbour is  $B_{i,j+1}$  and its vertical neighbour  $B_{i+1,j}$ ; note that neighbouring windows overlap in a column or a row.

The condition is that  $B_{i,j}B_{i,j+1} = B_{i,j+1}B_{i,j}$  and  $B_{i,j}B_{i+1,j} = B_{i+1,j}B_{i,j}$  whenever the windows concerned lie inside the array, the product being the ordinary matrix product.

$a(n)$  is the number of such arrays. The entry's offset is 1, so its term of index  $n$  is the count for  $n$  rows.

## 3 The count is a walk count

A  $2 \times 2$  window and its vertical neighbour together occupy 3 consecutive rows, so taking as state the last 2 rows makes every condition decidable one row at a time: appending a row produces the next window, and the two windows in hand are exactly a window and its vertical neighbour.

An array of  $n'$  rows is then a walk of length  $n'$  from the empty state, every array arising from exactly one walk, so  $a(n) = w^T M^{n'} f$  and the count is C-finite.

## 4 The degree bound, derived

The reachable part of the digraph has 502 states, 502 after trimming; identifying states with equal numbers of completions of every length, and then the mirror identification on the edges coming in, leaves  $S = 53$  blocks. The count satisfies the linear recurrence given by the characteristic polynomial of that  $53 \times 53$  matrix; the bound is computed, not assumed.

## 5 The decision procedure

Let  $c_1, \dots, c_D$  be the coefficients of a claimed recurrence  $a(n) = \sum_{i=1}^D c_i a(n-i)$ , let  $q(t) = t^D - \sum_{i=1}^D c_i t^{D-i}$  be its characteristic polynomial, and put

$$u_m = \tilde{w}^T L^{m-1} q(L) \tilde{f} \quad (m \geq 1).$$

Expanding the powers of  $L$  and using the displayed formula for  $a$ , one gets  $u_m = a(n) - \sum_{i=1}^D c_i a(n-i)$  with  $n = m + D$ . So  $u_m$  is exactly the residual of the claim at that index, and the recurrence holds at an index  $n$  if and only if the corresponding  $u_m$  vanishes.

Now  $u_m = \tilde{w}^T L^{m-1} y$  with  $y = q(L) \tilde{f}$  a fixed vector, so  $(u_m)$  satisfies the characteristic polynomial of  $L$ , of degree 53: writing that polynomial as  $t^{53} - \sum_{i=1}^{53} \lambda_i t^{53-i}$  gives  $u_{m+53} = \sum_{i=1}^{53} \lambda_i u_{m+53-i}$ . Hence 53 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector  $L^{m-1}y$ . The matrix  $M$  is never formed.

## 6 Results

**Theorem 1.** *For A186460, the recurrence  $a(n) = 2a(n-1) - 2a(n-3) + 2a(n-4) - a(n-5)$  holds for every  $n > 13$ , and fails at  $n = 13$ .*

*Proof.* The residuals  $u_m$  were computed exactly for  $m = 1, \dots, 95$ . The last nonvanishing one is at  $m = 8$ , that is at  $n = 13$ . After it, more than  $S = 53$  consecutive residuals vanish, so by Section 5 every later residual vanishes as well.  $\square$

The entry claims the recurrence for  $n > 13$ . The theorem is at least as strong.

**Theorem 2.** *For A186460, the generating function is exactly  $\sum_{n \geq 1} a(n)x^n = \frac{245x - 198x^2 + 6x^3 + 255x^4 - 115x^5 - 144x^6 + 36x^7}{1 - 2x + x^2}$ .*

*Proof.* Let  $N(x)$  and  $D(x)$  be the numerator and denominator above,  $D(0) = 1$ . The residual test applied to the recurrence read off  $D$  — namely  $a(n) = \sum_i c_i a(n-i)$  with  $c_i$  the negated coefficients of  $D$  — gives vanishing residuals for every  $n > 13$ , so the power series  $F(x) = \sum_{n \geq 1} a(n)x^n$  satisfies  $[x^k](F(x)D(x)) = 0$  for every  $k > 14$ . The remaining coefficients of  $F(x)D(x)$ , for  $k \leq 21$ , were computed from the walk count and agree with  $N(x)$  term by term. Hence  $F(x)D(x) = N(x)$  identically, which is the assertion.  $\square$

## 7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 31 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 245, 292, 590, 945, 1681, 2377, 4121, 5633, ...
- **An independent enumeration.** For  $n = 1, \dots, 2$  every one of the  $2^{8n}$  arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound  $S = 53$  over the whole vector, in exact integer arithmetic, never sampled and never truncated.

## 8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry [A186460](#), revision 9, last modified Apr 18 2018.

2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.
4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.